Can This Agent Even Do That? A Decidable Goal-Achievability Type Discipline for LLM-Synthesized Agent Skills

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Abstract

Modern autonomous agents rely on natural-language skills that specify tools, workflows, and desired outcomes. Such skills may be written by human experts or generated with LLM assistance, but their goals may be structurally impossible to achieve: a required tool may be unavailable, a constraint may be unsatisfiable, or agents may deadlock. We present a *skill compiler* that checks goal achievability before execution, without relying on an LLM judge. It combines capability-based reachability from automated planning with direct checking of multiparty communication protocols [1, 2]. An LLM may compact a skill into a formal pack of capabilities, preconditions, effects, goals, and protocol structure, which is then checked deterministically. A Coq-verified abstraction theorem guarantees that an *impossible* verdict is correct for the formal pack being checked. On a deterministic corpus of 15 skills, the checker correctly identifies all 6 achievable and all 7 genuinely impossible cases; the two false *achievable* results are deliberately constructed cases outside the static guarantee. The full evaluation takes less than one second. The checker uses no LLM tokens, while compacting the corpus costs 12.2K tokens compared with approximately 1.07M tokens for one round of corresponding agent executions, an approximately $87\times$ reduction. These results show that a lightweight static check can identify structurally impossible skills before execution at low cost.

1 Introduction

A modern agent is no longer merely a chatbot, but an autonomous system capable of taking actions beyond text generation—for example, writing code, sending emails, or executing trades. Such behaviour is often guided by natural-language artifacts that specify instructions, workflows, and constraints for the agent [3, 4, 6, 7]. A *skill* is one common way of expressing such guidance. It is a folder containing instructions and tooling that an agent can discover and load as needed [5], thereby *harnessing* a general-purpose agent toward a particular outcome.

There is growing evidence that this guidance matters. Across seven state-of-the-art open-source multi-agent systems, failure rates range from 41% to 86.7% [8]. More importantly, some of these failures can be addressed without changing the underlying model. In one system, simply adding a task-objective verification step to the specification improves task success by 15.6% [8, Appendix H]. Likewise, repairing the harness based on execution traces improves task completion by 6.3 to 18.4 percentage points across four agent benchmarks [4]. These results show that the instructions and structure surrounding an agent can matter as much as the model itself.

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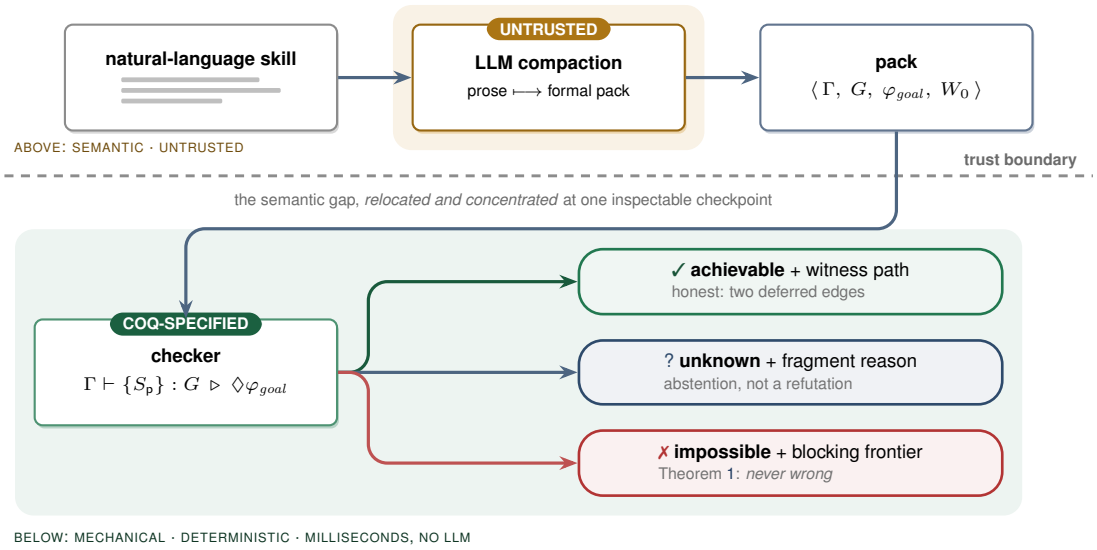

Figure 1: The trust boundary. An untrusted LLM compacts natural-language instructions into a formal pack. A deterministic checker then decides achievability over the pack using a Coq-verified abstraction theorem. The verdicts are asymmetric: *impossible* is sound (Theorem 1), while *achievable* depends on the accuracy of the pack (Section 3.1).

The question, however, is how we can know whether such guidance is sufficient to achieve the outcome it declares. A plan may *book a flight and email a confirmation* even though no email tool is available. It may pursue *a flight under $500* even though the available booking tool can return only premium fares. Or a planner may wait for a worker's result while the worker waits for the planner's answer, with each expecting a message that the other will never send. These are not merely execution failures. In each case, the goal is already unreachable before execution begins.

Such failures are structural and recurring, yet today the only way to discover them is often to run the skill and inspect what went wrong afterwards [4]. By then, the agent may already have spent substantial time and tokens taking actions, and a multi-agent system may already have become stuck waiting for communication that will never occur. A simple static check could, in principle, identify these problems before execution.

This paper asks a deliberately narrow question: **given a skill, can its declared goal be achieved at all?** We answer this question with a deterministic formal typing system. We do not verify that every step is safe or that the agent will behave correctly on every possible input. Instead, we check *achievability*. The check is tolerant of small details, detours, and payload variation, which should not by themselves cause a false alarm. At the same time, it is *sound for refutation*: when the compiler reports that a goal is impossible, that verdict is correct.

1.1 The idea, and the trust boundary

The architecture, shown in Figure 1, has two parts separated by a trust boundary. First, an LLM performs *compaction*: it reads the natural-language skill and turns it into a formal *pack* containing capabilities with preconditions and effects, a goal-marked global protocol, and an initial state. This step is *untrusted*: the LLM may misunderstand the skill or omit relevant details.

A deterministic *checker* then decides achievability using only the resulting pack. The checker is mechanically proved sound: if it returns *impossible*, then the goal is impossible under the formal pack it received. If the LLM omits a constraint or capability from the original skill, the checker cannot recover information that is not present in the pack; it will instead reason about the incomplete pack. Conversely, if the LLM introduces a constraint that was not in the original skill, the checker may report the resulting goal as impossible. Thus, our guarantee is about the checker's verdict with respect to the formal pack, not about whether the pack perfectly represents the original natural-language skill.

This does not make the compaction step a weakness that we simply ignore. The formal pack provides a structured target for the LLM to extract. Rather than asking the LLM to freely invent a formal model, we give it a predefined logical structure: capabilities have preconditions and effects, goals are explicit, and the protocol follows a fixed syntax. This makes compaction a constrained translation task and gives us a concrete artifact that can be inspected before execution.

Crucially, the gap between *what the user meant* and *what was formalized* cannot be eliminated. We instead make this gap explicit at the compaction step, where the resulting pack can be inspected and checked before execution. When the compaction is accurate, the checker can identify goals that are impossible before the agent spends time and tokens executing the skill. When the compaction is inaccurate, the checker still provides a sound verdict for the formal pack it received. The key design idea is therefore to move achievability checking from runtime execution to an explicit, inspectable checkpoint before execution.

1.2 Contributions

1. **A goal-achievability type discipline** (Section 3 and Appendices A–C) that combines capability-guarded reachability from automated planning with deadlock-freedom from multiparty session types. The discipline checks whole session configurations directly against a goal-marked global type synthesized from the skill. Building on the multiparty session type tradition of Honda, Yoshida, and Carbone [12, 13], and the direct session-checking approach of SMPS [1, 2], it adds capabilities, explicit goals, and a refutation-sound but achievement-incomplete analysis for LLM-drafted skills.
2. **A mechanized soundness specification** (Appendix D) formalized in Coq. We prove a refutation-sound abstraction theorem parameterized by simulation hypotheses, together with tolerance soundness and capability monotonicity: adding capabilities cannot make an achievable goal impossible. We also establish operational correspondence through subject reduction and session fidelity for the direct judgment over a labelled transition system. For single-label communication, we mechanize the cases where other participants continue to act concurrently; the world-changing action cases are proved on paper under explicit side conditions.
3. **A termination boundary** (Appendix D). When the set of participants is fixed, the symbolic search always terminates, and an *impossible* verdict is sound. Exact achievability over unbounded integers is undecidable, so the checker works on a widened abstraction rather than deciding the exact judgment. Likewise, allowing an unbounded number of new participants to be added during execution can make the problem undecidable, because the search is no longer guaranteed to remain finite.
4. **An implementation and evaluation** (Sections 4–5). We implement the checker together with an LLM-compaction front-end and a schema gate, and evaluate it on a corpus covering the main failure modes. The results are consistent with the theoretical guarantees: we observe no false impossibility verdicts, while every false achievability verdict falls into one of the two deferred cases.

2 Overview by Example

Consider a one-agent skill whose goal is *a flight is booked and a confirmation is sent*. The LLM compacts the skill into a formal pack. If no email tool is available, there is no action that can establish confirmation_sent.

$$\varphi_{goal} \;=\; \mathsf{booked} \wedge \mathsf{confirmation\_sent}$$

$$\Gamma(a) \;=\; \langle\, pre_a,\ \langle\, Add_a,\ Del_a,\ Asg_a,\ Nd_a \,\rangle \rangle$$

$$\begin{aligned}
\mathsf{search} &\mapsto \langle\, \top, && \langle\, \{\mathsf{searched}\}, && \emptyset, \emptyset, \emptyset \,\rangle \rangle \\
\mathsf{filter} &\mapsto \langle\, \mathsf{searched}, && \langle\, \{\mathsf{filtered}\}, && \emptyset, \emptyset, \emptyset \,\rangle \rangle \\
\mathsf{book} &\mapsto \langle\, \mathsf{filtered}, && \langle\, \{\mathsf{booked}\}, && \emptyset, \emptyset, \emptyset \,\rangle \rangle \\
\mathsf{email} &\mapsto \langle\, \mathsf{booked}, && \langle\, \{\mathsf{confirmation_sent}\}, && \emptyset, \emptyset, \emptyset \,\rangle \rangle
\end{aligned}$$

$$\mathrm{dom}(\Gamma_A) = \{\mathsf{search}, \mathsf{filter}, \mathsf{book}\} \qquad \mathrm{dom}(\Gamma_B) = \mathrm{dom}(\Gamma_A) \cup \{\mathsf{email}\}$$

Under STRIPS [17] frame semantics, the checker therefore returns *impossible* and identifies the missing capability. If the email tool is available, the goal is reachable through the sequence search ·

filter · book · email, which the checker returns as a witness. The same example is mechanized in Appendix D (the instance `FlightInstance`). Importantly, the booking itself remains reachable in the first case; the refutation concerns only the missing capability needed to complete the goal.

Tolerance is also important. A skill should not be rejected simply because an agent sends extra status messages, takes a clarifying step, or uses a different payload. Our analysis therefore allows these variations in three ways: *existential may-reachability*, where one successful path is sufficient; *session subtyping*, which allows safe interface variation; and *goal-relevant abstraction*, which ignores payload details that do not affect the goal. These mechanisms are defined in Appendices A–C.

3 Verification Model and Signals

Declared pack. The checker consumes $P = \langle \Gamma, G, \varphi_{goal}, W_0 \rangle$. Here, Γ contains capability declarations with guards and STRIPS-style effects; G is a goal-marked global protocol (a *global type*; Appendix A); φ_{goal} is the logical goal; and W_0 is the initial abstract world state, described by a constraint $init \in \mathcal{L}$. Any world satisfying this constraint is a possible initial world, written as $W_0 \vDash init$. The LLM produces P from the natural-language skill. A deterministic schema gate checks that the pack stays within the part of the language the checker can decide: a finite, fixed set of roles, guarded recursion, finite-domain booleans, and quantifier-free linear integer arithmetic. The formal syntax and semantics are given in Appendices A–B.

Verification signals. The checker uses three static signals:

Signal	Role	Output
tool availability and effects	capability reachability	missing capability or witness action
coordination structure	global protocol conformance	protocol mismatch or conformant protocol
goal and cost refinements	logical reachability	blocked guard or model

These signals are checked before execution. Runtime payload checks remain separate obligations and are not assumed by the static checker.

Decision procedure. After the schema gate, the checker (i) rejects undeclared capabilities and goal conditions that cannot be achieved by any available capability; (ii) checks role behaviors against G, requiring exact sender choices while allowing safe receiver-side branch supersets; and (iii) explores existential protocol/world successors from W_0, using Z3 [18] for guards and refinements. A goal condition that cannot be satisfied blocks the search. If the solver times out or the pack falls outside these restrictions, the checker returns UNKNOWN rather than *impossible*. The checker therefore returns:

$$\begin{array}{ll} \textsc{Impossible}(F) & \text{a sound explanation of why the goal cannot be reached,} \\ \textsc{Achievable}(\pi, O) & \text{a witness path } \pi \text{ and deferred obligations } O, \\ \textsc{Unknown}(r) & \text{an abstention with a reason } r. \end{array}$$

Each verdict records the pack digest, checker semantics, and artifact version.

Guarantees. Let a configuration consist of a protocol state and an abstract world state, and let abs map concrete configurations to abstract configurations. Assume step and goal simulation for the declared pack (Appendix D). If no abstract run reaches φ_{goal}, no concrete run represented by that pack reaches the concrete goal (Theorem 1). The analysis remains sound when the abstraction is made less detailed, and adding capabilities cannot turn an achievable goal into an *impossible* one (Theorems 2–3). Direct conformance satisfies subject reduction and session fidelity (Theorems 4–5). The Coq development covers single-label communication with concurrent actions by other participants; interleaving two world-changing actions requires an explicit commutativity assumption. The restrictions above guarantee termination of the search: *impossible* is sound for the pack, while *achievable* is a statement about the abstract search (Proposition 1). Allowing an unbounded dynamic addition of participants is an explicit undecidable extension, for which the implementation returns UNKNOWN. Full rules, assumptions, proofs, and mechanization scope appear in Appendix D.

3.1 Structured feedback and human oversight

The checker returns an explanation for IMPOSSIBLE and a witness path for ACHIEVABLE. These outputs provide concrete feedback rather than a single score. A bounded repair adapter may ask the LLM to fix a protocol, but it cannot invent capabilities or weaken the goal. Every revision is checked again by the deterministic checker. Human review remains necessary for the gap between the original intent and the formal pack. Automated prompt or scaffold search is an application of these verdicts, not an empirical contribution of this paper.

4 Implementation

We implemented the checker and its supporting components in Python. The compaction front-end sends the skill to an LLM and passes the result through a deterministic *schema gate* before the checker processes it. Packs that do not satisfy the schema are rejected at this stage. The checker implements the abstractions from Appendices A–C: STRIPS frame semantics for the world, existential search over the protocol, and Z3 for guard and goal satisfiability and arithmetic refinements. On success it returns a witness path; on refutation, it reports the reason, such as a missing capability, unsatisfiable guard, unachievable goal condition, or non-projectable handoff. The optional LLM front-end may use a NON_PROJECTABLE counterexample for one bounded repair round. It may not invent capabilities or weaken the goal, and the deterministic checker remains the final decision maker.

The conformance step ($G \vdash (\mathbb{M}; W)$, Section C.2) is implemented by means of a simple *type inference algorithm* since all typing rules are structural in $(\mathbb{M}; W)$. At each step we freely choose either one or two participant processes to reduce or a goal satisfied by the world obtaining the set of sessions which must be typed as premises of the typing rule.

The adapter also checks the participant side conditions of T-COMM/T-ACT/T-GOAL: it computes $\mathsf{prt}(G)$ and compares it with the roles for which the pack declares behaviour. A declared role outside $\mathsf{prt}(G)$ has contract End, so a non-trivial behavior for that role cannot conform. If the pack declares no behavior for a participant, the checker reports the missing role instead of leaving the assumption implicit.

5 Evaluation

We assembled a corpus of 15 skills covering the main failure modes: hallucinated planning, retry-forever loops, coordination deadlock, and refinement violations. The corpus also includes achievable controls with detours and informed branches, plus two deliberately *spurious* cases for the deferred checks. Each skill has a ground-truth semantic label; the positive class is *achievable*. Evaluation is deterministic and CPU-only: it requires no training or live model inference, completes the 15-case matrix in under one second on a commodity laptop, and gives each individual Z3 query a 10-second timeout.

	predicted achievable	predicted impossible
truly achievable	$\mathrm{TP} = 6$	$\mathrm{FN} = 0$
truly impossible	$\mathrm{FP} = 2$	$\mathrm{TN} = 7$

Two claims, both confirmed. **Soundness**: $\mathrm{FN} = 0$ — no achievable goal was wrongly refuted, consistent with Theorem 1 (achievability recall $6/6 = 100\%$). **Incompleteness, contained**: $\mathrm{FP} = 2$, and both are the planted spurious cases — one a false payload post-condition (Layer-C obligation), and one an under-specified goal (intent edge). No structural failure was missed in this proof-of-concept corpus. This is evidence of coverage, not a proof that these are the only possible failure modes. Each refutation reason matches its intended failure mode:

Failure mode (source)	**Verdict reason**
hallucinated planning — invokes a non-existent tool	MISSING_CAPABILITY
hallucinated planning — no establisher for a goal conjunct	GOAL_UNSAT
retry-forever loop — guard never satisfiable	BLOCKED_GUARD
coordination deadlock — unobserved choice / bad handoff	NON_PROJECTABLE
refinement violation — budget unsatisfiable on every run	GOAL_UNSAT

What the check costs. The checker and deterministic front-end use *zero* LLM tokens. The LLM is used once to compact each skill *version*, while a full agent run uses tokens on every *invocation*. Under the conservative model in Appendix E, compacting the 15-spec corpus costs 12.2K tokens and avoids $\sim$1.07M tokens per round of invocations, giving $\sim 87\times$ leverage. For every failure mode, the cost of compaction is recovered by the first prevented run. For a skill that is *not* broken, the check costs 2.9% of one successful run, or nothing.

A separate six-case suite tests behavior not covered by the main matrix: tail-recursive retry and non-terminating control, adding new participants during execution, refutation that remains valid when new participants are added, and conformant versus non-conformant declared role behavior. It yields two ACHIEVABLE, three IMPOSSIBLE, and one UNKNOWN. The abstention is reported separately rather than counted as a false negative because UNKNOWN makes no refutation claim. Together the suites contain 21 declared packs; the 15-case matrix remains the only binary ground-truth matrix.

6 Related Work

Multiparty session types [13, 16, 9] provide formal methods for checking communication protocols, including deadlock-freedom through projection and subtyping [14, 10]. Our protocol follows the synchronous formulation of this line of work [9–11]. We also follow the recent approach of checking a whole session configuration directly against a global type [1, 2], rather than first projecting the global type to local types. The new contribution is the world and goal layer: capability guards, effects, goal observation, and the asymmetric refutation guarantee are not modelled by classical MPST or new SMPS. These are not part of classical MPST [13, 16, 9] or the recent direct-checking formulation [1, 2].

Automated planning [17] provides methods for deciding whether a goal can be reached from an initial state using available actions. We use the same basic idea of preconditions and effects, but apply it to agent skills and combine it with multiparty communication protocols. The resulting checker asks whether the declared goal can be achieved given both the available capabilities and the communication protocol.

Recent work on agent verification addresses related problems but has different goals. *Provable coordination via message sequence charts* [20] uses formal communication protocols for LLM agents and focuses on coordination correctness rather than goal achievability. Its runtime-planning extension also studies LLM-generated coordination protocols. *VeriGuard* [21] separates offline policy verification from online monitoring and uses the result as a safety mechanism. *Agent behavioural contracts* [22] provide probabilistic, runtime-enforced compliance and composition conditions for multi-agent systems. In contrast, our checker makes a static achievability decision and guarantees that an *impossible* verdict is correct for the formal pack.

Skill-bundle scanners and verified-skill catalogs address the same deployment object: portable agent skills. NVIDIA's SkillSpector [23] and Verified Agent Skills pipeline [24] provide checks for skill bundles before deployment, including code, metadata, dependencies, and other security-related properties. These checks are complementary to our approach. They can inspect the original `SKILL.md` and its supporting files, while our checker operates on the formal pack produced from the skill. Their purpose is to identify security and consistency problems; ours is to determine whether the declared goal can be achieved with the available capabilities and protocol.

To our knowledge, the work above does not combine capability-based goal reachability with multiparty protocol checking for agent skills, nor provide the same refutation-sound guarantee for an *impossible* verdict.

7 Limitations and Future Work

- **Dependence on the declared pack.** The guarantees apply to the declared Γ and S. If the LLM incorrectly describes a capability or omits an important detail from the skill, the checker can only reason about the resulting pack. A practical system should therefore also inspect `SKILL.md`, scripts, dependencies, manifests, and permission metadata before running the achievability check.

- **Adding participants during execution.** The current procedure terminates when the set of participants is fixed. When new participants can be added during execution, the procedure may return UNKNOWN rather than make an unsupported decision (Proposition 2).
- **Abstraction.** A coarse α allows more behaviors to be treated as equivalent. This preserves the refutation guarantee of Theorem 1, but may make the checker more likely to return *achievable*.
- **Formalization gaps.** The Coq development covers the generic refutation-sound abstraction theorem and concrete examples, head-move action safety, and subject reduction/session fidelity with full bystander interleaving for the single-label communication model, all axiom-free. The executable checker is schema-gated and tested against the formal specification, but its exact symbolic state is not yet fully mechanized (Lemma 3 is on paper). The Coq model also does not yet combine participant matching, multi-branch choice, and world-changing interleaving in one development. Future work includes completing this mechanization, connecting interleaved session runs to the reachability search, proving equivalence between the declarative judgment and the projection-based adapter, and proving that abstract witnesses correspond to concrete sessions. The 21-pack evaluation is also only a proof of concept.

Broader impact. Pre-execution checking can reduce wasted computation (Appendix E) and prevent structurally impossible plans from reaching execution. However, an incorrect pack can create false confidence, for example by omitting a tool effect or weakening the user's actual goal. We therefore report the formal pack, the result of the check, and any remaining conditions that still require attention. Human review and runtime checks are still needed for consequential use. This work does not evaluate fairness, privacy, or deployment safety.

8 Conclusion

Agent skills provide instructions, workflows, and tools that guide autonomous agents toward particular outcomes. Whether a skill is written by a human expert or generated with the help of an LLM, a basic question remains: **can the declared goal be achieved at all?** Our work answers this question before execution, using a deterministic checker over a formal pack derived from the skill. The checker combines capability-based reachability with a directly checked global protocol, and its *impossible* verdict is sound for the formal pack it receives.

This guarantee does not remove the gap between the original skill and its formal representation. An incorrect or incomplete pack can lead to an incorrect *achievable* result, which is why the result must be understood relative to the declared pack. Nevertheless, when the pack is correct, the check can identify structural failures before the agent spends time and tokens executing the skill. This provides a simple and inspectable way to check whether an agent skill is achievable before execution.

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A Syntax

We use disjoint sets of predicates $p \in \mathsf{Pred}$ (boolean), numeric variables $x \in \mathsf{Var}$, roles $\mathsf{p}, \mathsf{q}, \mathsf{r} \in \mathcal{R}$, capability names $a \in \mathsf{Cap}$, and message labels $\ell \in \mathsf{Lab}$. Following the modern presentation of multiparty sessions [1, 2] we define the model from the bottom up: state and capabilities (Sections A.1–A.2), then *processes* (Section A.3), and finally the global *type* they are checked against directly (Section A.4).

In each grammar, "::=" lists the possible forms of an expression. In each inference rule below, the statement below the line holds when all premises above the line hold. Rule names identify the corresponding checker obligation; they are not additional assumptions.

A.1 State logic

$$\begin{array}{rcll} e & ::= & x \mid n \mid e+e \mid e-e \mid c \cdot e & (c,n \in \mathbb{Z}) \\ \varphi & ::= & p \mid \top \mid \bot \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid e \bowtie e & \bowtie \in \{<,\leq,=,\geq,>,\neq\} \end{array}$$

The state logic $\mathcal{L}$ is quantifier-free linear integer arithmetic [19] with uninterpreted boolean predicates — a decidable fragment for which entailment $\vDash$ and satisfiability are discharged by a satisfiability-modulo-theories (SMT) solver.

A.2 Worlds and capabilities

A *world* $W = \langle B, N \rangle$ assigns truth values to predicates $B : \mathsf{Pred} \to \mathbb{B}$ and numeric variables $N : \mathsf{Var} \to \mathbb{Z}$. A *capability context* Γ maps each available tool a to a guarded effect:

$$\Gamma(a) = \langle \mathit{pre}_a, \mathit{eff}_a \rangle, \qquad \mathit{pre}_a \in \mathcal{L},$$
$$\mathit{eff}_a = \langle \mathit{Add}_a \subseteq \mathsf{Pred},\ \mathit{Del}_a \subseteq \mathsf{Pred},\ \mathit{Asg}_a : \mathsf{Var} \rightharpoonup e,\ \mathit{Nd}_a : \mathsf{Var} \rightharpoonup \mathcal{L} \rangle$$

$$a \notin \mathrm{dom}(\Gamma) \iff \text{the agent does } \textit{not} \text{ possess tool } a \quad \text{(no hallucinated tools)}.$$

$\mathit{Add}/\mathit{Del}$ are the STRIPS add- and delete-lists (well-formedness requires $\mathit{Add}_a \cap \mathit{Del}_a = \emptyset$, so the order of union and difference in apply below is immaterial); Asg gives deterministic numeric updates $x := e$; Nd gives *nondeterministic* updates $x := *$ constrained by a formula over the new value (used to model post-conditions such as *"books a fare under 500"*).

A.3 Processes and multiparty sessions

Skills *execute* as processes; a multi-agent run is a session of named processes. We use the synchronous multiparty session calculus in its modern coinductive presentation [1, 2]: processes coinductively defined are possibly infinite *regular* trees (finitely many distinct subtrees), and labels are goal-relevant tokens, $\ell = \alpha(m)$ for concrete messages m — the tolerance dial α is a parameter of the label alphabet.

$$P \ ::=^{\mathit{coind}}\ \mathbf{0} \mid \mathsf{p}!\{\ell_i.\, P_i\}_{i\in I} \mid \mathsf{p}?\{\ell_i.\, P_i\}_{i\in I} \mid a.\, P \qquad\qquad \mathbb{M} \ = \ \mathsf{p}_1[P_1] \parallel \cdots \parallel \mathsf{p}_n[P_n]$$

with I finite and non-empty, labels ℓ_i pairwise distinct, and role names pairwise distinct. The output $\mathsf{p}!\{\ell_i.P_i\}$ *internally* chooses a label and sends it to p; the input $\mathsf{p}?\{\ell_i.P_i\}$ offers an *external* choice; $a.P$ invokes capability a — the only construct that changes the world. A skill S_{p} *is* a process: the behaviour declared for role p.

A.4 Global types

The single communication construct of a global type is a *directed* labelled choice — selection by p and branching at q in one construct. (A partnerless "p selects" admits no coherent view for the roles that did not choose; directed choice is the standard MPST construct [13, 9].)

$$G \ ::=^{\mathit{coind}}\ \mathsf{p}\to\mathsf{q} : \{\ell_i.\, G_i\}_{i\in I} \mid a@\mathsf{p}.\, G \mid \checkmark\varphi.\, G \mid \mathsf{End}$$

subject to $\mathsf{p} \neq \mathsf{q}$ and the same regularity and label conditions as processes. This is the *only* type in the discipline: Section C.2 types a whole session directly against G — there is no local-type grammar, no projection function, and no separate subtyping relation. Goal markers $\checkmark\varphi$ live *only* in G: achievability is decided on the global type (Rule ACH in C.3), and markers are consumed at typing time by T-GOAL without ever burdening a process. Only $a@\mathsf{p}$ changes the world; messages carry coordination information only.

B Operational Semantics

B.1 World transitions

A capability fires when its guard holds, producing a successor world. Because Nd is nondeterministic, $\langle W \rangle\, a\, \langle W' \rangle$ may relate one world to many. The relation is implicitly indexed by Γ: the notation $\mathit{pre}_a, \mathit{eff}_a$ is defined only when $\Gamma(a) = \langle \mathit{pre}_a, \mathit{eff}_a \rangle$, so every firing presupposes $a \in \mathrm{dom}(\Gamma)$.

$$\frac{W \vDash \mathit{pre}_a \qquad W' \in \mathrm{apply}(\mathit{eff}_a, W)}{\langle W \rangle\ a\ \langle W' \rangle}\ \textsc{World-Act}$$

$$\mathrm{apply}(\mathit{eff}_a, \langle B, N\rangle) = \langle B', N'\rangle, \qquad B' = (B \cup \mathit{Add}_a) \setminus \mathit{Del}_a$$

$$N'(x) = \begin{cases} [\![\mathit{Asg}_a(x)]\!]_N & \text{if } x \in \mathrm{dom}(\mathit{Asg}_a) \\ v \text{ with } \langle B, N[x \mapsto v]\rangle \vDash \mathit{Nd}_a(x) & \text{if } x \in \mathrm{dom}(\mathit{Nd}_a) \\ N(x) & \text{otherwise (frame)} \end{cases}$$

$[\![e]\!]_N$ is the usual integer evaluation of e, reading each variable from N. When several variables are in $\mathrm{dom}(\mathit{Nd}_a)$, each constraint is evaluated with only its own variable updated; the schema gate rejects an $\mathit{Nd}_a(x)$ that mentions another updated variable.

B.2 A labelled semantics for sessions and global types

Only capabilities touch the world, so sessions reduce as configurations $(\mathbb{M}; W)$; communication is synchronous [10]. Each transition carries a label Λ recording its *participants*, so that independent moves by disjoint roles can be identified. Two auxiliary maps drive the labelled system: the *participants* $\mathsf{prt}(\cdot)$ of a global type, a label, or a session, and the *capabilities* $\mathsf{cap}(\cdot)$ (the set of moves a global type can perform). On global types,

$$\mathsf{prt}(\mathsf{p}{\to}\mathsf{q} : \{\ell_i.\, G_i\}_{i\in I}) = \{\mathsf{p},\mathsf{q}\} \cup \textstyle\bigcup_{i\in I} \mathsf{prt}(G_i) \qquad \mathsf{prt}(\checkmark\varphi.G) = \mathsf{prt}(G)$$
$$\mathsf{prt}(a@\mathsf{p}.G) = \{\mathsf{p}\} \cup \mathsf{prt}(G) \qquad \mathsf{prt}(\mathsf{End}) = \emptyset,$$

on labels $\mathsf{prt}(\mathsf{p}\,\ell\,\mathsf{q}) = \{\mathsf{p},\mathsf{q}\}$, $\mathsf{prt}(a@\mathsf{p}) = \{\mathsf{p}\}$, $\mathsf{prt}(\checkmark\varphi) = \emptyset$, and on sessions $\mathsf{prt}(\mathbb{M}) = \{\mathsf{p} \mid \mathbb{M} \equiv \mathsf{p}[P] \,\&\, P \neq \mathbf{0}\}$ (the roles with a residual behaviour). The capabilities of a global type are

$$\mathsf{cap}(\mathsf{p}{\to}\mathsf{q} : \{\ell_i.\, G_i\}_{i\in I}) = \{\mathsf{p}\,\ell_i\,\mathsf{q} \mid i \in I\} \cup \textstyle\bigcup_{i\in I} \mathsf{cap}(G_i)$$
$$\mathsf{cap}(a@\mathsf{p}.G) = \{a@\mathsf{p}\} \cup \mathsf{cap}(G) \qquad \mathsf{cap}(\checkmark\varphi.G) = \{\checkmark\varphi\} \cup \mathsf{cap}(G) \qquad \mathsf{cap}(\mathsf{End}) = \emptyset.$$

The capability names used by a protocol are

$$\mathsf{caps}(G) = \{a \mid \exists \mathsf{p}.\ a@\mathsf{p} \in \mathsf{cap}(G)\}.$$

The LTS of sessions is defined by

S-Comm
$$\frac{k \in I \subseteq J}{(\mathsf{p}[\mathsf{q}!\{\ell_i.P_i\}_{i\in I}] \parallel \mathsf{q}[\mathsf{p}?\{\ell_j.Q_j\}_{j\in J}] \parallel \mathbb{M}; W) \xrightarrow{\mathsf{p}\,\ell_k\,\mathsf{q}} (\mathsf{p}[P_k] \parallel \mathsf{q}[Q_k] \parallel \mathbb{M}; W)}$$

S-Act
$$\frac{\langle W\rangle\ a\ \langle W'\rangle}{(\mathsf{p}[a.P] \parallel \mathbb{M}; W) \xrightarrow{a@\mathsf{p}} (\mathsf{p}[P] \parallel \mathbb{M}; W')}$$

where $\parallel$ is commutative, associative and $\mathsf{p}[\mathbf{0}] \parallel \mathbb{M} \equiv \mathbb{M}$. A non-terminated configuration with no successor is *stuck*; the deadlocking planner/worker handoff of Section 1 is the stuck two-role configuration in which both processes begin with an input. A goal marker is *not* carried by any process, and goal satisfaction is *not* a session transition: it is observed by the predicate $\mathit{goal}_{\varphi_{goal}}$ below. Session labels Λ are therefore communications and actions only, as in the Coq development. On the type side, marker discharge in the correspondence is fused with the next step (G-Goal-F below).

The checker relates a session to its protocol through a *labelled* transition system on global types, $(G, W) \rightsquigarrow_\Lambda (G', W')$, built from *head* rules that consume the leading construct and *interleaving*

rules that let a move by participants *disjoint* from the head commute inward:

$$\text{G-Comm-E}\quad \frac{k \in I}{(\mathsf{p}{\to}\mathsf{q}{:}\{\ell_i.G_i\}_{i\in I},\, W) \rightsquigarrow_{\mathsf{p}\,\ell_k\,\mathsf{q}} (G_k,\, W)} \qquad \text{G-Act-E}\quad \frac{\langle W\rangle\, a\, \langle W'\rangle}{(a@\mathsf{p}.\, G,\, W) \rightsquigarrow_{a@\mathsf{p}} (G,\, W')}$$

$$\text{G-Goal-E}\quad \frac{W \vDash \varphi}{(\checkmark\varphi.\, G,\, W) \rightsquigarrow_{\checkmark\varphi} (G,\, W)}$$

$$\text{G-Comm-I}\quad \frac{\forall i \in I.\ \big((G_i, W) \rightsquigarrow_\Lambda (G_i', W') \ \wedge\ \Lambda \in \mathsf{cap}(G_i)\big) \qquad \mathsf{prt}(\Lambda) \cap \{\mathsf{p},\mathsf{q}\} = \emptyset}{(\mathsf{p}{\to}\mathsf{q}{:}\{\ell_i.G_i\}_{i\in I},\, W) \rightsquigarrow_\Lambda (\mathsf{p}{\to}\mathsf{q}{:}\{\ell_i.G_i'\}_{i\in I},\, W')}$$

$$\text{G-Act-I}\quad \frac{(G, W) \rightsquigarrow_\Lambda (G', W') \qquad \Lambda \in \mathsf{cap}(G) \qquad \mathsf{prt}(\Lambda) \cap \{\mathsf{p}\} = \emptyset}{(a@\mathsf{p}.\, G,\, W) \rightsquigarrow_\Lambda (a@\mathsf{p}.\, G',\, W')}$$

$$\text{G-Goal-F}\quad \frac{W \vDash \varphi \qquad (G, W) \rightsquigarrow_\Lambda (G', W')}{(\checkmark\varphi.\, G,\, W) \rightsquigarrow_\Lambda (G',\, W')}$$

In G-Comm-I all branches take the same Λ to the same W'; for a communication $W' = W$, the only case the correspondence needs. G-Goal-E discharges a goal marker when $W \vDash \varphi$; it belongs to the head-only reduction used for achievability, where an unsatisfied marker is a checkpoint and blocks the path. In the labelled correspondence the goal rule is G-Goal-F: it discharges the marker and performs the continuation step in one transition — the same fused rule as `GS_Goal` in the Coq development. The marker is checked at the world where the step happens, so no goal-stability hypothesis is needed, and $\checkmark\varphi$ is not a label of the correspondence. The extraction (head) rules G-Comm-E/G-Act-E/G-Goal-E give the Λ-erased reduction $(G, W) \rightsquigarrow (G', W')$ that the achievability judgment below uses; the interleaving rules matter only for the operational correspondence of Section D.3, and their $\Lambda \in \mathsf{cap}(G)$ premise confines a commuting move to a capability the residual protocol actually offers. G-Comm-E is legitimate because choice is directed — the branch is a communication both endpoints observe, and Section C.2 checks that every third party can follow. A well-formed pack uses its declared goal φ_{goal} at each goal marker. A configuration is *goal-satisfying* when control is at such a marker or at the terminus and the world entails that declared goal:

$$goal_{\varphi_{goal}}(\checkmark\varphi_{goal}.G, W) \iff W \vDash \varphi_{goal}, \qquad goal_{\varphi_{goal}}(\mathsf{End}, W) \iff W \vDash \varphi_{goal},$$

$$\Gamma; G \vDash \Diamond_{init}\varphi_{goal} \iff \exists W_0 \vDash init.\ \exists (G', W').\ (G, W_0) \rightsquigarrow^* (G', W') \ \wedge\ goal_{\varphi_{goal}}(G', W').$$

The diamond is *existential*: a single witnessing run suffices — the formal source of detour-tolerance.

B.3 Realizability and subtyping, without projection

The classical route to ruling out a deadlocking handoff is *projection*: compute each role's local type $G{\upharpoonright}\mathsf{r}$ by structural recursion on G, taking the *merge* of a choice's branches at every role that neither sends nor receives, and declaring G *realizable* exactly when this partial function is defined everywhere — undefinedness of the merge is the static signature of a role forced to behave differently in branches it cannot distinguish. Session subtyping $\leq$ is then a second, separate coinductive relation on local types, letting a role's actual behaviour offer more external choices and fewer internal ones than its projected contract demands [14, 10, 11].

Both devices exist to answer questions about the *network as a whole* — "can every role tell the branches apart it needs to?", "may this implementation stand in for that contract?" — while type-checking only one role's syntax at a time. Section C.2 answers the same two questions with a single judgment, *typing the whole configuration directly against* G: a choice node's rule quantifies over every branch the protocol declares and requires the *same* residual configuration to check against each one. This is the conservative plain-merge condition, obtained without ever computing $\sqcap$ or a local type. The receiver-side subtyping direction (a receiver may accept a superset of the protocol's labels) is folded into the same rule as a label-set inclusion $I \subseteq J$. Tolerance therefore still has the same

three independent knobs from Section 2 — the existential $\Diamond$ (Section B.2), subtyping (now inside T-COMM, Section C.2), and the granularity of α — "flexible, tolerant of small details" is still the profile $\langle$coarse α, $\Diamond$, linear $\mathcal{L}\rangle$; a later safety layer flips the same machinery to the universal $\Box$ over permission predicates.

C The Achievability Type System

Achievability now factors into three premises, each checked by a separate terminating procedure (over the widened symbolic worlds of Theorem 6), combined by the top rule ACH: no hallucinated capability, direct conformance of the declared skills to G, and reachability of the goal.

C.1 Capability and goal typing

$$\text{T-EFF}\ \frac{\Gamma(a) = \langle \mathit{pre}, \mathit{eff}\rangle \qquad \varphi \vDash \mathit{pre}}{\Gamma \vdash \{\varphi\}\, a\, \{\mathrm{sp}(\varphi, \mathit{eff})\}} \qquad \text{T-MISS}\ \frac{a \notin \mathrm{dom}(\Gamma)}{\Gamma \vdash a \triangleright \text{✗}\ (\text{MISSING\_CAPABILITY})}$$

$\mathrm{sp}(\varphi, \mathit{eff})$ is the strongest postcondition of the STRIPS effect. T-MISS fires on hallucinated planning: the plan names a tool the context does not grant, and achievability is refuted without any search.

C.2 Direct conformance typing

Skills are typed *directly* against the global protocol and the world, by the coinductive judgment $G \vdash (\mathbb{M}; W)$ — read "G types session $\mathbb{M}$ starting from world W." There is no local type, no projection, and no separate subtyping relation: the receiver-side subtyping direction and the unobserved-choice check are structural side conditions of one rule.

$$\text{T-END}\ \frac{}{\mathsf{End} \vdash (\mathsf{p}[\mathbf{0}]; W)}$$

$$\text{T-ACT}\ \frac{\exists W'.\ \langle W\rangle\, a\, \langle W'\rangle \qquad \forall W'.\ \langle W\rangle\, a\, \langle W'\rangle \implies G \vdash (\mathsf{p}[P] \parallel \mathbb{M}; W') \qquad \mathsf{prt}(G) \cup \{\mathsf{p}\} = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p}\}}{a@\mathsf{p}.\, G \vdash (\mathsf{p}[a.P] \parallel \mathbb{M}; W)}$$

$$\text{T-GOAL}\ \frac{G \vdash (\mathbb{M}; W) \qquad W \vDash \varphi \qquad \mathsf{prt}(G) = \mathsf{prt}(\mathbb{M})}{\checkmark\varphi.\, G \vdash (\mathbb{M}; W)}$$

$$\text{T-COMM}\ \frac{G_i \vdash (\mathsf{p}[P_i] \parallel \mathsf{q}[Q_i] \parallel \mathbb{M}; W) \qquad \forall i \in I \subseteq J \qquad \mathsf{prt}(\mathsf{p}{\to}\mathsf{q} : \{\ell_i.G_i\}_{i\in I}) = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p}, \mathsf{q}\}}{\mathsf{p}{\to}\mathsf{q} : \{\ell_i.G_i\}_{i\in I} \vdash (\mathsf{p}[\mathsf{q}!\{\ell_i.P_i\}_{i\in I}] \parallel \mathsf{q}[\mathsf{p}?\{\ell_j.Q_j\}_{j\in J}] \parallel \mathbb{M}; W)}$$

T-ACT pairs type and world in lockstep with WORLD-ACT: the unconsumed type $a@\mathsf{p}.G$ sits with the *pre*-effect world W and the residual G with the *post*-effect world W' — the same convention as G-ACT-E (Section B.2). The second premise quantifies over *every* successor world: an *Nd* effect is nondeterministic, the session may land in any successor, and subject reduction needs the residual typed at each. The first premise keeps the guard requirement. For capabilities without *Nd* the successor is unique and the rule reads as before. In T-COMM the sender p offers exactly the protocol's branch set I, while the receiver q may accept *more* ($I \subseteq J$) — the one subtyping direction the rule keeps (safe interface slack at the receiver). Requiring *every* protocol branch ℓ_i to be checked against the *same* bystanders $\mathbb{M}$ is the plain-merge form of the unobserved-choice condition, so a role forced to behave differently across branches it cannot distinguish makes no derivation possible — the rule fails closed, and deadlock-freedom falls out of T-COMM itself with no merge to compute (Theorem 4). Each rule additionally carries a *participant-matching* side condition tying the protocol's participants to the session's: $\mathsf{prt}(\mathsf{p}{\to}\mathsf{q} : \{\ell_i.G_i\}) = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p}, \mathsf{q}\}$ in T-COMM, $\mathsf{prt}(G) \cup \{\mathsf{p}\} = \mathsf{prt}(\mathbb{M}) \cup \{\mathsf{p}\}$ in T-ACT, and $\mathsf{prt}(G) = \mathsf{prt}(\mathbb{M})$ in T-GOAL. These make the invariant "the roles the protocol still mentions are exactly the roles still running" hold at every node, ruling out a session that carries a stray participant the protocol has forgotten (or vice versa) — the well-formedness that lets the interleaving

cases of Section D.3 match a bystander move on the session to a $\mathsf{cap}(\cdot)$-labelled move on the type. T-END closes the discipline at the terminus: the terminated protocol End types a single idle role $\mathsf{p}[\mathbf{0}]$ — and, through the congruence $\mathsf{p}[\mathbf{0}] \parallel \mathbb{M} \equiv \mathbb{M}$, any session all of whose roles have terminated — the base case of the participant invariant, since $\mathsf{prt}(\mathsf{End}) = \emptyset$ and an idle role contributes nothing to $\mathsf{prt}(\mathbb{M})$. T-ACT is derivable only when $\langle W \rangle\, a\, \langle W' \rangle$ holds, which itself presupposes $a \in \mathrm{dom}(\Gamma)$: this is where hallucinated capabilities die, with T-MISS as the diagnostic.

C.3 The achievability judgment

$$\textsc{Ach}\quad \frac{\mathsf{caps}(G) \subseteq \mathrm{dom}(\Gamma) \qquad \exists W_0 \vDash \mathit{init}.\ G \vdash (\mathsf{p}_1[S_{\mathsf{p}_1}] \parallel \cdots \parallel \mathsf{p}_n[S_{\mathsf{p}_n}]; W_0) \ \wedge\ \Gamma; (G, W_0) \vDash \Diamond\varphi_{goal}}{\Gamma \vdash \{S_{\mathsf{p}}\}_{\mathsf{p}}\ :\ G \triangleright \Diamond\varphi_{goal}}$$

where $\{\mathsf{p}_1, \ldots, \mathsf{p}_n\} = \mathsf{prt}(G)$: the judgment takes one declared skill S_{p} per role of the protocol. Read top-to-bottom: no hallucinated tools; some plausible initial world exists from which the declared skills directly conform to the protocol (deadlock-free, every capability call guarded) *and* that same world reaches the goal. The reachability half is unchanged from Section B.2 and decided on Γ, G alone, before any S_{p} is known — exactly what lets the checker run on the LLM-compacted pack the moment it exists; conformance is the additional, separately checked premise once the skills are declared, and needs no transport lemma: the receiver-side subtyping slack is already inside T-COMM.

C.4 Reachability rules

The diamond is derived by the following inductive system, realized by the checker as a finite forward search over $\rightsquigarrow$; unlike T-COMM/T-ACT/ T-GOAL, it never mentions a session $\mathbb{M}$, which is what makes it checkable on the pack alone.

$$\textsc{Reach-Goal}\quad \frac{W \vDash \varphi_{goal} \qquad G = \mathsf{End} \ \vee\ \exists G'.\, G = \checkmark\varphi_{goal}.G'}{\Gamma; (G, W) \vDash \Diamond\varphi_{goal}}$$

$$\textsc{Reach-Step}\quad \frac{(G, W) \rightsquigarrow (G', W') \qquad \Gamma; (G', W') \vDash \Diamond\varphi_{goal}}{\Gamma; (G, W) \vDash \Diamond\varphi_{goal}} \qquad \textsc{Reach-Or}\quad \frac{\exists k \in I.\, \Gamma; (G_k, W) \vDash \Diamond\varphi_{goal}}{\Gamma; (\mathsf{p}{\to}\mathsf{q}{:}\{\ell_i.G_i\}_{i\in I}, W) \vDash \Diamond\varphi_{goal}}$$

REACH-OR is derivable from REACH-STEP with G-COMM-E; we keep it because the search implements branching this way.

D Metatheory

We separate what is *mechanized in Coq* (Theorems 1–3; Theorems 4–5 for the single-label communication fragment with full interleaving; the head-move action case; and the concrete instances, all axiom-free) from what is *proved on paper* (the action-interleaving cases, Lemma 3, termination, and the undecidability boundary).

D.1 Soundness of the abstraction

Let $\mathcal{C}$ be the concrete configuration space of protocol/world pairs, $\mathcal{A}$ the abstract configuration space the checker explores, and $abs : \mathcal{C} \to \mathcal{A}$ the abstraction. We require:

$$\textbf{(step-sim)}\ \ C \rightsquigarrow C' \Rightarrow abs(C)\,\hat{\rightsquigarrow}\,abs(C'), \qquad \textbf{(goal-sim)}\ \ goal_{\varphi_{goal}}(C) \Rightarrow \hat{goal}(abs(C)).$$

Lemma 1 (Transport)**.** *If $C_0 \rightsquigarrow^* C$ then $abs(C_0)\,\hat{\rightsquigarrow}^*\,abs(C)$.*

Proof. By induction on the reflexive-transitive closure. The empty run is reflexivity. For $C_0 \rightsquigarrow^* C' \rightsquigarrow C$, the induction hypothesis gives $abs(C_0)\hat{\rightsquigarrow}^* abs(C')$ and STEP-SIM gives $abs(C')\hat{\rightsquigarrow}abs(C)$; compose them. Mechanized as `reach_abs`, parameterized by STEP-SIM. □

Lemma 2 (Reachability monotonicity)**.** *If every step of $\rightsquigarrow_1$ is a step of $\rightsquigarrow_2$ (that is, $x \rightsquigarrow_1 y$ implies $x \rightsquigarrow_2 y$), then $s \rightsquigarrow_1^* w$ implies $s \rightsquigarrow_2^* w$.*

Proof. By induction on the closure $s \rightsquigarrow_1^* w$. The empty run transports by reflexivity. For $s \rightsquigarrow_1^* u \rightsquigarrow_1 w$, the induction hypothesis gives $s \rightsquigarrow_2^* u$; the hypothesis turns the last step $u \rightsquigarrow_1 w$ into $u \rightsquigarrow_2 w$; appending it gives $s \rightsquigarrow_2^* w$. Mechanized as `reach_mono`. □

D.2 Refutation soundness, tolerance, monotonicity

Theorem 1 (Refutation soundness)**.** *If the abstract system has no reachable goal state from $abs(G, W_0)$, then no declared run of that pack configuration reaches φ_{goal}. Hence an* impossible *verdict is never wrong relative to the pack, provided* STEP-SIM *and* GOAL-SIM *hold.*

Proof. We prove the contrapositive: if *some* declared run reaches the goal, then the abstract system reaches an abstract goal state. Suppose $C_0 \rightsquigarrow^* C$ with $goal_{\varphi_{goal}}(C)$. By Lemma 1, $abs(C_0) \hat{\rightsquigarrow}^* abs(C)$, and by GOAL-SIM, $\hat{goal}(abs(C))$. Thus $abs(C)$ is an abstract goal state reachable from $abs(C_0)$, contradicting the hypothesis. Hence no declared run reaches φ_{goal}: a verdict of *impossible* — read off exactly the premise, that the abstract search exhausts without hitting an abstract goal — is never wrong. Mechanized, axiom-free as a theorem schema parameterized by the two simulation hypotheses (the existential witness is $abs(C)$): □

```
Theorem refutation_sound :  forall s0,
(~ exists a, reach astep (abs s0) a /\ agoal a) ->
(~ exists w, reach cstep s0 w /\ cgoal w).
(* Print Assumptions refutation_sound.  => Closed under the global
context *)
```

Lemma 3 (The symbolic abstraction satisfies the hypotheses)**.** *Let $abs(G, \langle B, N\rangle) = (G, \langle B, \psi\rangle)$ for any accumulated constraint $\psi \in \mathcal{L}$ with $N \vDash \psi$. Let an abstract step apply the same Add/Del update to B and replace ψ by the strongest postcondition of the numeric effect, and let $\hat{goal}(G, \langle B, \psi\rangle)$ hold when goal holds at control G and $\psi \wedge \varphi_{goal}$ is satisfiable under B. Then* STEP-SIM *and* GOAL-SIM *hold, and the widening $\psi \mapsto \top$ preserves both.*

Proof. GOAL-SIM: if $\langle B, N\rangle \vDash \varphi_{goal}$ and $N \vDash \psi$, then N itself witnesses satisfiability of $\psi \wedge \varphi_{goal}$ under B. STEP-SIM: a concrete firing needs $\langle B, N\rangle \vDash pre_a$, so $\psi \wedge pre_a$ is satisfiable (N witnesses it) and the symbolic edge exists; the boolean update is identical on both sides, and the concrete successor satisfies the strongest postcondition by definition. Widening weakens ψ, which only adds abstract states and edges, so both properties are preserved (Theorem 2). Hence Theorem 1 applies to the search the implementation runs. This lemma is on paper; mechanizing the symbolic state is future work (Section 7). □

Theorem 2 (Tolerance soundness)**.** *If a* coarser *over-approximation (more edges) cannot reach the goal, neither can any finer system. Hence coarsening α to tolerate more detail never produces a false refutation.*

Proof. Let $\rightsquigarrow_{\text{fine}}$ and $\rightsquigarrow_{\text{coarse}}$ be the two abstract step relations, with the coarsening an *over*-approximation: every fine step is also a coarse step, $x \rightsquigarrow_{\text{fine}} y \Rightarrow x \rightsquigarrow_{\text{coarse}} y$. Suppose, for contradiction, that the finer system reaches the goal: $s_0 \rightsquigarrow_{\text{fine}}^* a$ with $\hat{goal}(a)$. By Lemma 2 (with $\rightsquigarrow_1 = \rightsquigarrow_{\text{fine}}$, $\rightsquigarrow_2 = \rightsquigarrow_{\text{coarse}}$), $s_0 \rightsquigarrow_{\text{coarse}}^* a$; and $\hat{goal}(a)$ still holds. So the coarser system reaches a goal state, contradicting the hypothesis. Hence no coarser-refuted goal is reachable in any finer system: coarsening α only *adds* abstract edges, so it can never turn an unreachable goal into a reachable one — refutation is preserved. Mechanized as `tolerance_sound`, axiom-free. □

Theorem 3 (Capability monotonicity)**.** *If $\Gamma \subseteq \Gamma'$ and Γ reaches φ_{goal}, then Γ' reaches φ_{goal}. Granting tools never makes an achievable goal impossible.*

Proof. Enlarging the capability context only *adds* guarded effects: since $\Gamma \subseteq \Gamma'$, every capability firing available under Γ is available under Γ' with the same guard and effect (WORLD-ACT in B.1 quantifies over $a \in \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$), so every step of the Γ-transition relation is a step of the Γ'-relation. Suppose Γ reaches the goal configuration C from C_0. By Lemma 2, the same configuration is reachable under Γ', and $goal_{\varphi_{goal}}(C)$ is unchanged because the goal predicate does not depend on Γ. Hence Γ' reaches φ_{goal} via the same witnessing run. The other premises of ACH

are monotone in the same direction ($\mathsf{caps}(G) \subseteq \mathrm{dom}(\Gamma) \subseteq \mathrm{dom}(\Gamma')$ still holds, and realizability and conformance do not mention Γ beyond $a \in \mathrm{dom}(\Gamma)$), so the whole judgment is preserved. The transport half is mechanized as `cap_monotone`, axiom-free, stated over an inclusion of step relations; the reduction of $\Gamma \subseteq \Gamma'$ to that inclusion is the WORLD-ACT argument above, on paper. □

Concrete instance. The flight skill of Section 2, variant A, is mechanized as `FlightInstance`. We prove `flight_refuted` (no run reaches booked ∧ confirmation_sent, since no step touches conf) and `booking_reachable` (a booking is still reachable). Both are axiom-free, witnessing non-vacuity of Theorem 1.

D.3 Operational correspondence: subject reduction and session fidelity

The direct judgment $G \vdash (\mathbb{M}; W)$ is not merely preserved by reduction: the session and its protocol step *in lockstep* over the labelled transition systems of Section B.2. This is the standard soundness backbone of a session calculus [13, 11], obtained here *directly*, with no projection and no merge. For world-changing interleavings, the statements below assume one explicit frame condition: effects of participant-disjoint actions commute. Communication-only reductions satisfy it vacuously; goal markers need no condition, because G-GOAL-F checks the marker at the world where the step happens.

Lemma 4 (Residual capability)**.** *If* $(G, W) \rightsquigarrow_\Lambda (G', W')$*, then* $\Lambda \in \mathsf{cap}(G)$.

Lemma 5 (Participant agreement)**.** *If* $G \vdash (\mathbb{M}; W)$*, then* $\mathsf{prt}(G) = \mathsf{prt}(\mathbb{M})$.

Both follow by induction on the transition or typing derivation, respectively; they are the capability-membership and participant invariants used in the interleaving cases below.

Theorem 4 (Subject reduction)**.** *If* $G \vdash (\mathbb{M}; W)$ *and* $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$*, then* $(G, W) \rightsquigarrow_\Lambda (G', W')$ *for some* G' *with* $G' \vdash (\mathbb{M}'; W')$.

Theorem 5 (Session fidelity)**.** *If* $G \vdash (\mathbb{M}; W)$ *and* $(G, W) \rightsquigarrow_\Lambda (G', W')$*, then* $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$ *for some* $\mathbb{M}'$ *and* W' *with* $G' \vdash (\mathbb{M}'; W')$.

We prove both by induction on the typing derivation $G \vdash (\mathbb{M}; W)$, with a case analysis on the transition; the one auxiliary fact used throughout is that typing respects pointwise-equal sessions (`ctypes_ext`), which lets the interleaving cases reassociate independent updates.

Proof of Theorem 4 (subject reduction). By induction on $G \vdash (\mathbb{M}; W)$.

Case T-END ($G = \mathsf{End}$, every role idle). Then $\mathbb{M}$ has no output or pending action, so no session transition exists and the claim holds vacuously.

Case T-GOAL ($G = \checkmark\varphi.G_0$, with $W \vDash \varphi$ and $G_0 \vdash (\mathbb{M}; W)$, $\mathsf{prt}(G_0) = \mathsf{prt}(\mathbb{M})$). A marker is not a session construct, so the step $(\mathbb{M}, W) \to_\Lambda (\mathbb{M}', W')$ is a step of G_0's session; the induction hypothesis gives $(G_0, W) \rightsquigarrow_\Lambda (G', W')$ with $G' \vdash (\mathbb{M}'; W')$. Since $W \vDash \varphi$, rule G-GOAL-F discharges the marker and performs the same step, $(\checkmark\varphi.G_0, W) \rightsquigarrow_\Lambda (G', W')$; the residual typing is the one the induction hypothesis provides. No stability hypothesis is needed: the marker is checked at W, where the step happens.

Case T-COMM, with head $\mathsf{p}\to\mathsf{q}$ and branches G_i. Here $\mathbb{M}$ is $\mathsf{p}[\mathsf{q}!\dots] \parallel \mathsf{q}[\mathsf{p}?\dots] \parallel \mathbb{M}_0$, and each branch satisfies $G_i \vdash (\mathsf{p}[P_i] \parallel \mathsf{q}[Q_i] \parallel \mathbb{M}_0; W)$. The transition is a communication $\Lambda = \mathsf{r}\,\ell\,\mathsf{t}$. Because p offers only outputs to q and q only inputs from p, the sender/receiver shapes force a dichotomy.

- *Head move* ($\mathsf{r} = \mathsf{p}$). Then $\mathsf{t} = \mathsf{q}$ and $\ell = \ell_k \in I$, and $\mathbb{M}' = \mathsf{p}[P_k] \parallel \mathsf{q}[Q_k] \parallel \mathbb{M}_0$. Rule G-COMM-E gives $(G, W) \rightsquigarrow_{\mathsf{p}\ell_k\mathsf{q}} (G_k, W)$, and $G_k \vdash (\mathbb{M}'; W)$ is exactly the k-th premise.
- *Interleaving move* ($\mathsf{r} \neq \mathsf{p}$). The shapes then force $\mathsf{r}, \mathsf{t} \notin \{\mathsf{p}, \mathsf{q}\}$: r is an output so $\mathsf{r} \neq \mathsf{q}$, and t is an input so $\mathsf{t} \neq \mathsf{p}$, while $\mathsf{t} = \mathsf{q}$ would force $\mathsf{r} = \mathsf{p}$. Since r, t lie in the bystanders $\mathbb{M}_0$, the *same* communication is enabled in each branch's configuration $\mathsf{p}[P_i] \parallel \mathsf{q}[Q_i] \parallel \mathbb{M}_0$ (the head updates do not touch r, t). The induction hypothesis at each i yields G_i' with $(G_i, W) \rightsquigarrow_\Lambda (G_i', W)$ and $G_i' \vdash (\mathsf{p}[P_i] \parallel \mathsf{q}[Q_i] \parallel \mathbb{M}_0'; W)$, where $\mathbb{M}_0'$ is $\mathbb{M}_0$ after the bystander step. As $\mathsf{prt}(\Lambda) = \{\mathsf{r}, \mathsf{t}\}$ is disjoint from $\{\mathsf{p}, \mathsf{q}\}$, rule G-COMM-I assembles

$(G, W) \rightsquigarrow_\Lambda (\mathsf{p}{\to}\mathsf{q}{:}\{\ell_i.G_i'\}, W)$, and re-applying T-COMM (the sender/receiver at p, q are untouched, and each branch is retyped by the induction hypothesis, reassociating the independent updates via `ctypes_ext`) gives the residual typing.

Communications leave the world fixed, so no independence hypothesis is needed. For a world-changing action at the head (T-ACT), G-ACT-E matches and the residual G is typed at the *same* post-effect world the rule reaches — whichever successor world the session's firing picks, the universal premise of T-ACT types the residual at it. Interleaving *two* actions requires their effects to commute ($\mathsf{eff}_a \circ \mathsf{eff}_b = \mathsf{eff}_b \circ \mathsf{eff}_a$ for disjoint a, b), a frame-independence side condition that participant-disjointness alone does not supply; under it the T-ACT interleaving case goes through identically. □

Proof of Theorem 5 (session fidelity). By induction on $G \vdash (\mathbb{M}; W)$, inverting the global step $(G, W) \rightsquigarrow_\Lambda (G', W')$. T-END admits no global step in the correspondence, so the claim holds vacuously. For T-GOAL, the step is G-GOAL-F; its premise $(G_0, W) \rightsquigarrow_\Lambda (G', W')$ is realised by the induction hypothesis as a session step, which is the required one — the marker exists only on the type side and is discharged by the same transition. For T-COMM, the global step is either G-COMM-E — and the prescribed head communication $\mathsf{p}\ \ell_k\ \mathsf{q}$ is enabled by S-COMM, landing in the k-th premise — or G-COMM-I with $\mathsf{prt}(\Lambda) \cap \{\mathsf{p}, \mathsf{q}\} = \emptyset$; then the induction hypothesis on any branch produces a bystander communication, whose endpoints lie outside $\{\mathsf{p}, \mathsf{q}\}$, so S-COMM fires it on $\mathbb{M}$ and T-COMM retypes the result. The action cases mirror those of Theorem 4. □

Mechanization. Both theorems are mechanized axiom-free for the single-label communication fragment — the fragment in which bystander interleaving is unconditional — in `DirectTypingSR.v`, over labelled session and global transition systems, with no projection, merge, or local type:

```
Theorem subject_reduction :  forall W G s s' L,
ctypes W G s -> lstep L s s' ->
exists G', gstep W L G G' /\ ctypes W G' s'.
Theorem session_fidelity :  forall W G G' s L,
ctypes W G s -> gstep W L G G' ->
exists s', lstep L s s' /\ ctypes W G' s'.
(* Print Assumptions => Closed under the global context (both).  *)
```

The head-move case for world-changing *actions* is additionally mechanized in `DirectTyping.v` (`type_directed_safety`, also `progress`). Goal labels appear in neither label alphabet: the paper's correspondence uses the same fused goal rule (G-GOAL-F) as the Coq `GS_Goal`. Participant-matching side conditions, labelled action interleaving, and multi-branch choice are still absent from the mechanization. Completing a faithful mechanization of the paper rules remains work in Section 7.

Concrete instance. `HandoffInstance` mechanizes the planner/worker handoff of Section 1 on both sides. The *good* pairing — planner sends, worker delivers, goal checked — is proved `ctypes`-typed (`good_typed`) and its run is proved to reach a world satisfying the goal (`good_runs_to_goal`). The *bad* pairing — both roles start with an input, exactly the deadlock of Section 1 — is proved `Stuck` (`bad_stuck`), and the contrapositive of `progress` then gives, for free, that *no non-trivial global type can ever type it* (`bad_untypeable`): the rule rejects the classical deadlocking handoff without any projection ever being computed.

D.4 Incompleteness, by design

Proposition 1 (Incompleteness)**.** $\Gamma; G \vDash \Diamond_{init} \varphi_{goal}$ *may hold while no concrete runtime execution reaches* φ_{goal}*: the witnessing abstract path can be spurious when* α *collapses a distinction that mattered. The system is* sound for ✗ and incomplete for ✓.

This is the price of decidable, tolerant over-approximation; tightening α trades tolerance for precision. The residue is confined to two edges, motivating the layered architecture:

- **Payload faithfulness** (bottom edge): a tool's declared post-condition ("filters to fares < 500") may be false — a runtime obligation for a deterministic monitor (Layer C), not a static one.

- **Intent fidelity** (top edge): the formalized goal may not capture what the user meant — irreducibly semantic, surfaced at the single compaction checkpoint for human review.

D.5 Decidability and the autonomy boundary

Theorem 6 (Termination and soundness of the widened search)**.** *In the fragment with finitely many roles (static topology), regular global types represented by finite tail-recursive control graphs, decidable state logic $\mathcal{L}$, and $\mathcal{L}$-definable effects, the widened symbolic search terminates on every pack. It decides the widened abstract judgment; for $\Gamma; G \vDash \Diamond_{init}\varphi_{goal}$ its* IMPOSSIBLE *answer is sound (Theorem 1 with Lemma 3) and its* ACHIEVABLE *answer may be spurious (Proposition 1).*

Proof. The pack-level achievability judgment $\Gamma; G \vDash \Diamond_{init}\varphi_{goal}$ (Section B.2) is an existential reachability query over configurations (G', W); we exhibit the terminating procedure and place its two answers.

The search space is finite. A configuration has two components. (i) *Control*, the residual global type G' reachable from G by $\rightsquigarrow$. Since G is a regular tree (tail-recursion: finitely many distinct subtrees, Section A.3) and each $\rightsquigarrow$ step either descends into a subtree (G-COMM-E, G-ACT-E) or erases a marker (G-GOAL-E), the set $\{G' : (G, \_) \rightsquigarrow^* (G', \_)\}$ is a subset of the finitely many subtrees of G, hence finite. (ii) *Symbolic world.* The boolean part is a valuation $B : \mathsf{Pred} \to \mathbb{B}$ over the finitely many predicates named in the pack, so there are $2^{|\mathsf{Pred}|}$ boolean states. The numeric part is summarized by an accumulated constraint $\psi \in \mathcal{L}$ (a conjunction of the guards taken and the Nd-postconditions assumed); on a control-graph back edge the reference procedure applies the widening $\psi \mapsto \top$, forgetting accumulated numeric constraints. This one-element numeric summary is an over-approximation and, together with the finite boolean valuations, leaves only finitely many symbolic worlds. The product with finite control is therefore finite.

Each edge is computable. Firing a capability requires checking $W \vDash pre_a$ and forming sp; testing goal-satisfaction requires $W \vDash \varphi$. Both are entailment/satisfiability queries in quantifier-free linear integer arithmetic with uninterpreted predicates — a decidable theory, discharged by an SMT solver. Branch selection (REACH-OR) is a finite disjunction. Existence of an initial world is decided by the same SMT query over $init$; each satisfying symbolic initial valuation seeds the finite search.

Termination and correctness. Forward search (breadth-first over $\rightsquigarrow$) maintains a visited set of (control, symbolic-world) pairs; because that set is finite it adds each pair at most once, so the search halts. It answers ACHIEVABLE iff it enumerates a goal-satisfying configuration, and IMPOSSIBLE otherwise. Soundness of the IMPOSSIBLE verdict is Theorem 1 with Lemma 3; and widening only *enlarges* the reachable set (it weakens constraints, adding edges), so by Theorem 2 it never introduces a false refutation. Hence the search always terminates and decides the widened abstract judgment; IMPOSSIBLE transfers to the exact judgment, ACHIEVABLE does not (Proposition 1). □

Why not exact decidability. With unbounded integers, updates $x := x \pm 1$, and guards $x = 0$, $x \geq 1$ under recursive control, packs encode two-counter machines, so the exact judgment $\Gamma; G \vDash \Diamond_{init}\varphi_{goal}$ is undecidable even with a fixed set of participants. The widening is what buys termination; the price is the spurious-ACHIEVABLE residue of Proposition 1.

Proposition 2 (Undecidability in a dynamic-topology extension)**.** *Consider the extension G^+ of the grammar with* add r. *G, which creates (adds) a new participant during execution: the new participant runs a declared finite behaviour template and brings one fresh numeric variable, initialized by the step that adds it. With an unbounded role population and dynamic channels, achievability in G^+ is undecidable — and no widening restores termination, since the reachable control space itself becomes infinite.*

Proof sketch. By reduction from the control-state reachability problem for communicating finite-state machines (CFSMs), which is undecidable [15]. Fix a CFSM instance: finite-control machines M_1, M_2 exchanging messages over unbounded FIFO channels, and a target control state q_f of M_1; deciding whether a run reaches q_f is undecidable.

We compute, from this instance, a skill whose goal is achievable iff the CFSM reaches q_f. Each machine M_i becomes a role whose process mirrors its control graph: a transition that sends message m creates, during execution, a new *courier* participant carrying m (this is exactly the ability to add a

new participant during execution that the proposition hypothesizes), and a transition that receives m synchronizes with the oldest outstanding courier for that channel, so the unbounded family of couriers realizes the unbounded FIFO. The required order is enforced through sequence numbers: the step that adds the courier writes the current send counter into the courier's fresh variable, and a receive is guarded on that variable matching the receive counter — this is where the per-courier fresh variable of G^+ is essential, since the fixed Var of the base fragment cannot order unboundedly many couriers. The Asg/Nd machinery of Section A.2 supplies the counter updates. A boolean predicate $\mathtt{at\_}q_f$ is established by the single capability guarding M_1's entry into q_f, and we set $\varphi_{goal} = \mathtt{at\_}q_f$. By construction a run of the skill reaches a state satisfying φ_{goal} iff the CFSM has a run reaching q_f: the courier discipline then puts the skill's reachable configurations in correspondence with the CFSM's channel contents and control states. The construction is a computable map from CFSM instances to packs, so a decision procedure for achievability would decide CFSM control-state reachability — impossible. Hence achievability is undecidable once new participants may be added during execution. (The finiteness argument of Theorem 6 breaks at exactly the point this reduction exploits: an unbounded number of couriers makes the set of reachable controls infinite.) □

The two results isolate one concrete boundary: with static finite topology the widened search is total and refutation-sound, while adding an unbounded number of new participants during execution defeats any such finite search. This is a boundary on topology, not a claim that all forms of agent autonomy are undecidable. The implementation returns UNKNOWN when a pack violates the static-topology side condition of Theorem 6, unless an independent capability or establisher refutation has already proved it impossible.

D.6 The partition of obligations

Corollary 1. *The architecture separates "will the goal be achieved" into four obligations on disjoint substrates:*

Concern	Where	Mechanism
goal achievability	static, terminating	Coq specification; deterministic search; no LLM
per-step safety / least privilege	Layer B (future)	same engine, □ + permission tiers
payload faithfulness	Layer C (future), runtime	deterministic monitors with blame
intent fidelity	top edge, synthesis	one human checkpoint

D.7 Verifiable feedback beyond a scalar reward

The verdict is a heterogeneous verification signal rather than a scalar score. An IMPOSSIBLE result identifies which premise failed and carries a blocking frontier; ACHIEVABLE carries a witness path and explicitly retains the two deferred obligations; UNKNOWN is an abstention outside the decidable fragment, not a refutation. These structured outcomes can serve as cheap negative labels for automated skill or prompt search without an LLM judge, while the intent-fidelity checkpoint keeps the irreducible semantic decision with a human. We treat such search as an application of the verifier, not as a contribution evaluated here.

E Token economics of pre-execution refutation

The practical case for deciding achievability *before* execution is an economic one, so we state it in tokens rather than adjectives. Two asymmetries carry it. First, no model sits in the trusted path: the schema gate, direct conformance and Z3 reachability spend **zero tokens**, and the deterministic front-end spends zero as well. Only the optional LLM compaction spends tokens, and it is paid *once per skill version*. Second, an agent turn re-sends its whole context: writing S for the harness prompt, K for the loaded skill, g for the mean tokens appended per turn and o for output per turn, a run of T turns reads $T(S{+}K) + g\,T(T{-}1)/2$ input and To output tokens. Compaction is linear in the skill and paid once; a doomed run is *quadratic* in its turns and recurs on every invocation.

Runtime waste is necessarily a model — it prices a run that, if the refutation is correct, never happens — so we publish the model rather than a single number. Each refutation reason carries

a (low, typical, high) band for how long an agent flails before that failure stops it, and how many agents are billed. With conservative defaults (S=2500, g=700, o=250) and a median real consumer skill ($K \approx 1.7$K tokens), compaction costs $\sim$2.8K tokens against the following avoided waste per prevented invocation:

Verdict reason	turns (lo–typ–hi)	waste/run	leverage
MISSING_CAPABILITY	3–8–14, 1 agent	50 240	18×
GOAL_UNSAT	8–18–30, 1 agent	176 040	63×
NON_CONFORMANT	6–15–30, 2 agents	261 900	94×
NON_PROJECTABLE	6–15–30, 2 agents	261 900	94×
BLOCKED_GUARD	12–25–50, 1 agent	305 750	110×

The ordering is the robust part, and it is the expected one. MISSING_CAPABILITY is cheapest at run time: the agent calls a tool that is not there and learns immediately, and most of the cost is the improvisation that follows. BLOCKED_GUARD is dearest, because a precondition no run can satisfy is precisely the case an agent cannot learn from — it retries to the harness turn cap. The two multi-party reasons bill two contexts. GOAL_UNSAT additionally carries a cost the token bill does not show: a run whose goal conjunct has no establisher can terminate *believing* it succeeded. DYNAMIC_TOPOLOGY claims no savings at all, because an abstention prevents nothing.

Over the 15-spec corpus the seven structural refutations cost 12.2K tokens of compaction and avoid $\sim$1.07M tokens per round of invocations (band 0.28M–3.15M): $\sim$87× leverage, or nothing at all on the deterministic path. Break-even therefore falls *inside* the first prevented run in every mode — between 0.9% and 5.5% of it. The honest denominator for a healthy skill, where the check buys nothing and still costs something, is 2.9% of one successful run over the corpus and 4.0% for a median real skill; with the deterministic front-end, zero. Prompt caching changes the price of the wasted tokens, not their number. Halving every waste default leaves leverage at 9×–55× and the ordering unchanged. The model, its defaults and their justification are in the artifact (`skillc.tokens`, `skillc cost`); the Coq development costs no tokens and is amortized over every skill ever checked.

NeurIPS Paper Checklist

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   Justification: The abstract and Section 1 state that refutation is sound relative to the declared pack, distinguish UNKNOWN from refutation, and identify the mechanized and paper-only proof fragments.

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   Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Sections 4–5 describe the deterministic checker, verdict semantics, corpus construction, binary accounting, and separate treatment of abstentions.

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